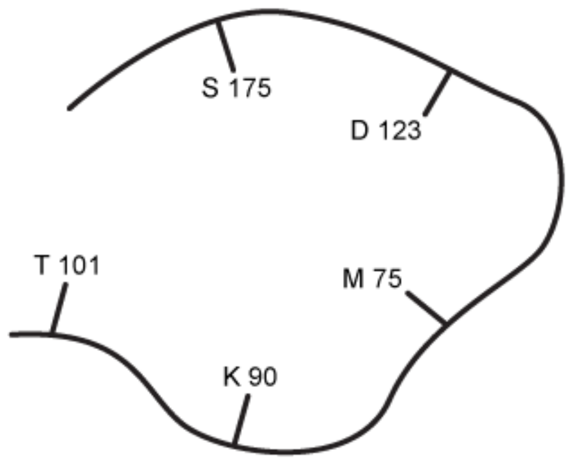
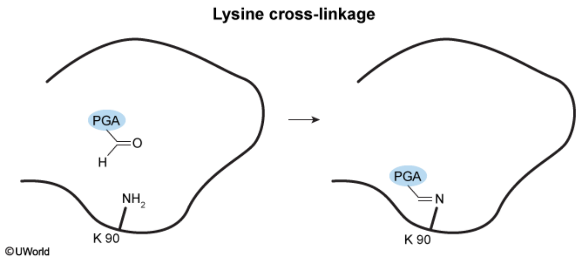
Polyglutaraldehyde (PGA) can form linkages with amines. PGA could potentially cross-link with which residue in the active site of signal peptidase, shown below?



- A. T 101 [4%]
- B. M 75 [5%]
- C. S 175 [6%]
- D. K 90 [83%]

Correct
83% answered correctly

Explanation:



Amino acids all have a similar backbone structure composed of a chiral carbon (α-carbon) bound to four groups: A **hydrogen atom** (–H), an **amino group** (–NH_3^+), a **carboxyl group** (–COO^-), and an **R group**. The R group is the **variable side chain** and is unique to each amino acid. Each R group has distinct physical and biochemical properties that influence reactivity with other compounds.

The question states that polyglutaraldehyde (PGA) can form linkages with amino groups. All free amino acids have amino groups in their backbone, but when an amino acid is incorporated into a protein, this backbone amino group is converted to an amide, which cannot cross-link with PGA. Therefore, PGA will be able to cross-link only with residues that have an amino group in the R group (side chain). Lysine (K) is the only standard amino acid with an amine in its side chain, so PGA will be able to **link only with K 90**.

(Choices A, B, and C) Methionine (M) has a thioether side chain (C–S–C); threonine (T) and serine (S) have hydroxyl groups (–OH) in their side chains. These amino acids do not have amines in their side chains and therefore cannot cross-link with PGA.

Educational objective:
All amino acids have a similar backbone structure that includes a chiral carbon (α-carbon) bound to four groups: A hydrogen atom (–H), an amino group ($–NH_3^+$), a carboxyl group ($–COO^-$), and an R group. The R group is a variable side chain that is unique to each amino acid and influences protein reactivity with other compounds.

Time spent:0 sec
Subject:
Biochemistry

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System:
Amino Acids and Proteins